Project 5: Healthcare System using Agile

1. Abstract

The Healthcare System project focuses on creating a digital ecosystem to manage patient records, doctor schedules, and clinical workflows. Developed under the **Agile methodology**, the project prioritizes patient safety and data privacy by delivering critical modules like "Electronic Health Records (EHR)" and "Appointment Scheduling" in early Sprints. The system aims to reduce administrative overhead and improve the quality of patient care by providing doctors with instant access to medical histories and diagnostic reports.

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2. Introduction

- **2.1 Introduction**: This project involves the development of a comprehensive Hospital Management System (HMS) that digitizes the daily operations of a medical facility.
- **2.2 Problem Identification**: Manual patient record-keeping is prone to errors, leads to delayed access during emergencies, and creates difficulties in tracking long-term treatment plans.
- **2.3 Need of the project**: There is a vital need for a secure, centralized system that bridges the gap between doctors, patients, and pharmacies while ensuring compliance with healthcare data regulations.
- **2.4 Project Scheduling**: The project is executed in four Sprints: Sprint 1 (Patient & Doctor Modules), Sprint 2 (Appointment & Billing), Sprint 3 (Pharmacy & Lab Integration), and Sprint 4 (Security & Compliance Testing).
- **2.5 Objectives**: To automate patient registration, manage doctor availability in real-time, and ensure secure storage of sensitive medical data.

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3. Software Requirement Specification

- **3.1 Purpose**: To provide an efficient and error-free platform for managing medical information and clinical services.

- **3.2 Scope**: Includes EHR management, doctor-patient portal, laboratory management, and billing.
- **3.3 Hardware/Software Requirement (minimum)**:
    - **Hardware**: Core i5 Processor, 16GB RAM (for database handling), 200GB SSD.
    - **Software**: Java JDK 17, MySQL Database, Windows/Linux OS.
- **3.4 Tools**: Eclipse IDE, Selenium WebDriver for functional testing, Maven, and Jira for Agile backlog management.
- **3.5 Software Process Model**: **Agile Scrum Model**, enabling clinicians to review modules frequently and ensure the system meets practical medical needs.

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4. System Design

- **4.1 Data Dictionary**: Defines critical data entities such as Patient_History, Prescription_ID, Doctor_Specialty, and Appointment_Slot.
- **4.2 ER Diagram**: Shows the complex relationships between Patient, Doctor, Appointment, Diagnosis, and Medicine.
- **4.3 DFD (Data Flow Diagram)**: Traces the flow of medical data from initial patient check-in to doctor consultation and pharmacy fulfillment.
- **4.4 Object Diagram**: Illustrates the state of specific medical objects at a given time, such as an active prescription or a confirmed surgery slot.

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5. Implementation

- **5.1 Program Code**: Developed using Java for high-performance backend processing and Selenium scripts to automate the validation of patient login and record retrieval.
- **5.2 Output Screens**: Snapshots of the Patient Dashboard, Doctor's Consultation Window, and Pharmacy Inventory View.

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6. Testing

- **6.1 Test Data**: Use of various patient age groups, complex medical history strings, and duplicate appointment scenarios.
- **6.2 Test Result**: Documented success in data encryption tests and validation of automated reminders for patient follow-ups.

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7. User Manual

- **7.1 How to use project guidelines**: Step-by-step instructions for medical staff on adding new patients and for patients on viewing test results.
- **7.2 Screen Layouts and Description**: Visual breakdown of the EHR interface, highlighting icons for "Vitals," "Allergies," and "Emergency Contact".

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8. Project Applications and Limitations

- **Applications**: Suitable for large hospitals, private clinics, and diagnostic centers.
- **Limitations**: Requires high-level security to prevent data breaches; system performance depends on high server availability during peak hours.

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9. Conclusion and Future Enhancement

- **Conclusion**: The Agile-based development allowed the system to remain flexible and patient-centric, significantly improving clinic efficiency.
- **Future Enhancement**: Integration of Telemedicine features (video calls), AI-based symptom checkers, and IoT connectivity for remote patient monitoring.

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10. Bibliography & References

- Referencing the Agile Manifesto, Selenium testing guides, and HIPAA compliance documentation.